

Artwork: Test Image1014

Scale-Invariant Coordinate System

$$\tilde{X} = 2 \left(\frac{x}{W} \right) - 1, \quad \tilde{Y} = 1 - 2 \left(\frac{y}{H} \right) \quad X = \frac{\left(\tilde{X} + \frac{8}{159} \right)}{\frac{78}{103}}, \quad Y = \frac{\left(\tilde{Y} - \frac{26}{197} \right)}{\frac{136}{149}}, \quad \theta = \arctan \left(\frac{Y}{X} \right)$$

Layer Definitions Boundary Manifolds

$$C_0(X, Y, v) = \frac{39 - 2v^2}{40} \cdot \left(1 + \frac{4}{155} Y \right)$$

$$X_1 = \left(X + \frac{15}{106} \right), \quad Y_1 = \left(Y - \frac{22}{41} \right)$$

$$\tau_1(X, Y) = -\frac{5}{148} \cos^2 \left(\frac{2736}{175} X_1 + \frac{1234}{195} Y_1 + \frac{10}{189} \cos \left(\frac{479}{128} X_1 + \frac{208}{95} Y_1 \right) \right) - \frac{11}{76} \cos^4 \left(\frac{655}{112} X_1 + \frac{2743}{133} Y_1 + \frac{5}{6} \cos \left(\frac{176}{65} X_1 + \frac{584}{133} Y_1 \right) \right) - \frac{3}{121} \cos^6 \left(\frac{4505}{162} X_1 + \frac{1174}{97} Y_1 - \frac{7}{1} \right)$$

$$X_2 = \left(X + \frac{8}{125} \right), \quad Y_2 = \left(Y + \frac{25}{198} \right)$$

$$\tau_2(X, Y) = \frac{43}{193} \cos^2 \left(\frac{1197}{74} X_2 + \frac{328}{57} Y_2 - \frac{33}{190} \cos \left(\frac{412}{105} X_2 + \frac{43}{22} Y_2 \right) \right) + \frac{32}{121} \cos^4 \left(\frac{351}{59} X_2 + \frac{2797}{141} Y_2 + \frac{10}{157} \cos \left(\frac{316}{147} X_2 + \frac{785}{156} Y_2 \right) \right) + \frac{47}{200} \cos^6 \left(\frac{3839}{137} X_2 + \frac{119}{10} Y_2 \right) +$$

$$X_3 = \left(X + \frac{11}{186} \right), \quad Y_3 = \left(Y + \frac{98}{183} \right)$$

$$\tau_3(X, Y) = \frac{32}{163} \cos^2 \left(\frac{2047}{129} X_3 + 6.002 Y_3 - \frac{16}{125} \cos(4.002 X_3 + 2 Y_3) \right) + \frac{29}{153} \cos^4 \left(\frac{43}{7} X_3 + \frac{2659}{133} Y_3 - \frac{25}{182} \cos \left(\frac{401}{200} X_3 + 5 Y_3 \right) \right) + \frac{39}{197} \cos^6 \left(\frac{619}{22} X_3 + 11.997 Y_3 + \frac{22}{157} \cos \right)$$

$$X_4 = \left(X - \frac{20}{141} \right), \quad Y_4 = \left(Y + \frac{100}{199} \right)$$

$$\tau_4(X, Y) = \frac{1}{59} \cos^2(16 X_4 + 6 Y_4) + \frac{3}{98} \cos^4(6 X_4 + 20 Y_4) + \frac{1}{84} \cos^6(28 X_4 + 12 Y_4) + \frac{4}{137} \cos^8(12 X_4 + 36 Y_4) C_4(X, Y, v) = \frac{25 + 21v + v^2}{40} \cdot \left(1 - \frac{1}{8} X_4 + \frac{1}{1} \right)$$

$$X_5 = \left(X - \frac{14}{107} \right), \quad Y_5 = \left(Y - \frac{67}{186} \right)$$

$$\tau_5(X, Y) = \frac{42}{181} \cos^2 \left(\frac{2347}{152} X_5 + \frac{285}{49} Y_5 - \frac{13}{126} \cos \left(\frac{835}{198} X_5 + \frac{79}{41} Y_5 \right) \right) - \frac{10}{61} \cos^4 \left(\frac{73}{13} X_5 + \frac{2189}{105} Y_5 - \frac{69}{190} \cos \left(\frac{519}{191} X_5 + \frac{107}{18} Y_5 \right) \right) - \frac{11}{147} \cos^6 \left(\frac{4556}{167} X_5 + \frac{997}{85} Y_5 + \frac{66}{113} \cos \right)$$

$$X_6 = \left(X - \frac{17}{151}\right), \quad Y_6 = \left(Y + \frac{37}{119}\right)$$

$$\tau_6(X, Y) = -\frac{39}{196} \cos^2 \left(\frac{1487}{93} X_6 + \frac{1067}{150} Y_6 - \frac{43}{57} \cos \left(\frac{437}{160} X_6 + \frac{478}{131} Y_6 \right) \right) - \frac{27}{190} \cos^4 \left(\frac{207}{34} X_6 + \frac{499}{26} Y_6 - \frac{66}{83} \cos \left(\frac{88}{107} X_6 + \frac{883}{185} Y_6 \right) \right) - \frac{19}{74} \cos^6 \left(\frac{2114}{73} X_6 + \frac{2101}{194} Y_6 - \frac{15}{18} \cos \left(\frac{437}{160} X_6 + \frac{478}{131} Y_6 \right) \right)$$

$$X_7 = \left(X + \frac{8}{37}\right), \quad Y_7 = \left(Y + \frac{61}{197}\right)$$

$$\tau_7(X, Y) = -\frac{7}{34} \cos^2 \left(\frac{2393}{150} X_7 + \frac{654}{113} Y_7 \right) + \frac{79}{192} \cos^4 \left(\frac{1389}{184} X_7 + \frac{3891}{197} Y_7 - \frac{93}{95} \cos \left(\frac{15}{22} X_7 + \frac{1109}{200} Y_7 \right) \right) - \frac{1}{34} \cos^6 \left(\frac{5149}{181} X_7 + \frac{2232}{173} Y_7 - \frac{9}{28} \cos \left(\frac{451}{81} X_7 + \frac{204}{101} Y_7 \right) \right)$$

$$X_8 = \left(X + \frac{5}{184}\right), \quad Y_8 = \left(Y - \frac{53}{99}\right)$$

$$\tau_8(X, Y) = \frac{25}{173} \cos^2 \left(\frac{3153}{197} X_8 + \frac{474}{77} Y_8 + \frac{7}{86} \cos \left(\frac{690}{187} X_8 + \frac{93}{53} Y_8 \right) \right) - \frac{4}{51} \cos^4 \left(\frac{294}{55} X_8 + \frac{1423}{74} Y_8 - \frac{47}{113} \cos \left(\frac{362}{191} X_8 + \frac{847}{173} Y_8 \right) \right) - \frac{24}{185} \cos^6 \left(\frac{5575}{197} X_8 + \frac{1947}{161} Y_8 - \frac{69}{131} \cos \left(\frac{690}{187} X_8 + \frac{93}{53} Y_8 \right) \right)$$

$$X_9 = \left(X - \frac{1}{25}\right), \quad Y_9 = \left(Y + \frac{26}{177}\right)$$

$$\tau_9(X, Y) = \frac{1}{120} \cos^2 \left(\frac{1573}{93} X_9 + \frac{621}{113} Y_9 + \frac{11}{31} \cos \left(\frac{1015}{193} X_9 + \frac{193}{85} Y_9 \right) \right) - \frac{61}{175} \cos^4 \left(\frac{533}{93} X_9 + \frac{2858}{157} Y_9 + \frac{245}{144} \cos \left(\frac{37}{49} X_9 + \frac{1137}{193} Y_9 \right) \right) + \frac{5}{128} \cos^6 \left(\frac{3869}{137} X_9 + \frac{2179}{187} Y_9 - \frac{11}{18} \cos \left(\frac{1015}{193} X_9 + \frac{193}{85} Y_9 \right) \right)$$

$$X_{10} = \left(X + \frac{31}{180}\right), \quad Y_{10} = \left(Y - \frac{52}{191}\right)$$

$$\tau_{10}(X, Y) = \frac{1}{47} \cos^6 \left(\frac{1031}{37} X_{10} + \frac{731}{60} Y_{10} + \frac{1}{6} \cos \left(\frac{710}{119} X_{10} + 3Y_{10} \right) \right) + \frac{1}{68} \cos^8 \left(\frac{792}{65} X_{10} + \frac{2400}{67} Y_{10} + \frac{35}{194} \cos \left(\frac{182}{61} X_{10} + 6.998Y_{10} \right) \right) - \frac{3}{79} \cos^{12} \left(\frac{4257}{89} X_{10} + \frac{2684}{133} Y_{10} - \frac{1}{6} \cos \left(\frac{710}{119} X_{10} + 3Y_{10} \right) \right)$$

$$X_{11} = \left(X + \frac{41}{171}\right), \quad Y_{11} = \left(Y - \frac{20}{73}\right)$$

$$\tau_{11}(X, Y) = \frac{22}{139} \cos^2 \left(\frac{491}{31} X_{11} + \frac{642}{109} Y_{11} + \frac{35}{153} \cos \left(\frac{351}{88} X_{11} + \frac{292}{147} Y_{11} \right) \right) + \frac{23}{131} \cos^4 \left(\frac{727}{125} X_{11} + \frac{3963}{200} Y_{11} - \frac{15}{82} \cos \left(\frac{292}{147} X_{11} + \frac{563}{113} Y_{11} \right) \right) + \frac{14}{115} \cos^6 \left(\frac{5147}{185} X_{11} + \frac{1133}{96} Y_{11} - \frac{15}{82} \cos \left(\frac{292}{147} X_{11} + \frac{563}{113} Y_{11} \right) \right)$$

$$X_{12} = \left(X + \frac{15}{76}\right), \quad Y_{12} = \left(Y - \frac{25}{108}\right)$$

$$\tau_{12}(X, Y) = -\frac{35}{176} \cos^2 \left(\frac{2420}{151} X_{12} + \frac{1039}{173} Y_{12} \right) - \frac{39}{196} \cos^4 \left(\frac{1090}{183} X_{12} + \frac{2797}{140} Y_{12} - \frac{11}{181} \cos(2X_{12} + 5Y_{12}) \right) - \frac{39}{196} \cos^6(28X_{12} + 12.002Y_{12}) - \frac{27}{136} \cos^8 \left(\frac{719}{60} X_{12} + \frac{377}{105} Y_{12} - \frac{11}{181} \cos(2X_{12} + 5Y_{12}) \right)$$

$$X_{13} = \left(X + \frac{5}{107}\right), \quad Y_{13} = \left(Y - \frac{53}{156}\right)$$

$$\tau_{13}(X, Y) = \frac{27}{92} \cos^2 \left(\frac{178}{11} X_{13} + \frac{491}{80} Y_{13} + \frac{85}{198} \cos \left(\frac{706}{177} X_{13} + \frac{367}{185} Y_{13} \right) \right) + \frac{9}{32} \cos^4 \left(\frac{992}{165} X_{13} + \frac{896}{45} Y_{13} - \frac{48}{137} \cos \left(\frac{201}{101} X_{13} + \frac{613}{123} Y_{13} \right) \right) + \frac{15}{94} \cos^6 \left(\frac{4703}{168} X_{13} + \frac{1245}{104} Y_{13} \right)$$

$$X_{14} = \left(X + \frac{59}{168}\right), \quad Y_{14} = \left(Y - \frac{43}{118}\right)$$

$$\tau_{14}(X, Y) = -\frac{106}{163} \cos^2 \left(\frac{2132}{125} X_{14} + \frac{91}{17} Y_{14} - \frac{199}{200} \cos \left(\frac{309}{80} X_{14} + \frac{431}{186} Y_{14} \right) \right) - \frac{5}{14} \cos^4 \left(\frac{644}{135} X_{14} + \frac{2097}{106} Y_{14} + \frac{298}{199} \cos \left(\frac{455}{151} X_{14} + \frac{142}{27} Y_{14} \right) \right) - \frac{111}{194} \cos^6 \left(\frac{847}{30} X_{14} + \frac{224}{185} Y_{14} \right)$$

$$X_{15} = \left(X + \frac{22}{199}\right), \quad Y_{15} = \left(Y - \frac{47}{146}\right)$$

$$\tau_{15}(X, Y) = -\frac{16}{145} \cos^2 \left(\frac{881}{55} X_{15} + \frac{1052}{157} Y_{15} + \frac{147}{190} \cos \left(\frac{217}{83} X_{15} + \frac{69}{146} Y_{15} \right) \right) + \frac{1}{64} \cos^4 \left(\frac{863}{152} X_{15} + \frac{3693}{179} Y_{15} + \frac{285}{166} \cos \left(\frac{391}{130} X_{15} + \frac{925}{179} Y_{15} \right) \right) - \frac{5}{136} \cos^6 \left(\frac{4781}{171} X_{15} + \frac{2}{5} Y_{15} \right)$$

Master Composition (Recursive Layer Synthesis)

$$H_v^{(0)}(X, Y) = C_0(X, Y, v)$$

$$H_v^1(X, Y) = (1 - W_1(X, Y)) \cdot H_v^0(X, Y) + W_1(X, Y) \cdot C_1(X, Y, v)$$

$$H_v^2(X, Y) = (1 - W_2(X, Y)) \cdot H_v^1(X, Y) + W_2(X, Y) \cdot C_2(X, Y, v)$$

$$H_v^3(X, Y) = (1 - W_3(X, Y)) \cdot H_v^2(X, Y) + W_3(X, Y) \cdot C_3(X, Y, v)$$

$$H_v^4(X, Y) = (1 - W_4(X, Y)) \cdot H_v^3(X, Y) + W_4(X, Y) \cdot C_4(X, Y, v)$$

$$H_v^5(X, Y) = (1 - W_5(X, Y)) \cdot H_v^4(X, Y) + W_5(X, Y) \cdot C_5(X, Y, v)$$

$$H_v^6(X, Y) = (1 - W_6(X, Y)) \cdot H_v^5(X, Y) + W_6(X, Y) \cdot C_6(X, Y, v)$$

$$H_v^7(X, Y) = (1 - W_7(X, Y)) \cdot H_v^6(X, Y) + W_7(X, Y) \cdot C_7(X, Y, v)$$

$$H_v^8(X, Y) = (1 - W_8(X, Y)) \cdot H_v^7(X, Y) + W_8(X, Y) \cdot C_8(X, Y, v)$$

$$H_v^9(X, Y) = (1 - W_9(X, Y)) \cdot H_v^8(X, Y) + W_9(X, Y) \cdot C_9(X, Y, v)$$

$$H_v^{10}(X, Y) = (1 - W_{10}(X, Y)) \cdot H_v^9(X, Y) + W_{10}(X, Y) \cdot C_{10}(X, Y, v)$$

$$H_v^{11}(X, Y) = (1 - W_{11}(X, Y)) \cdot H_v^{10}(X, Y) + W_{11}(X, Y) \cdot C_{11}(X, Y, v)$$

$$H_v^{12}(X, Y) = (1 - W_{12}(X, Y)) \cdot H_v^{11}(X, Y) + W_{12}(X, Y) \cdot C_{12}(X, Y, v)$$

$$H_v^{13}(X, Y) = (1 - W_{13}(X, Y)) \cdot H_v^{12}(X, Y) + W_{13}(X, Y) \cdot C_{13}(X, Y, v)$$

$$H_v^{14}(X, Y) = (1 - W_{14}(X, Y)) \cdot H_v^{13}(X, Y) + W_{14}(X, Y) \cdot C_{14}(X, Y, v)$$

$$H_v^{15}(X, Y) = (1 - W_{15}(X, Y)) \cdot H_v^{14}(X, Y) + W_{15}(X, Y) \cdot C_{15}(X, Y, v)$$

$$H_v(X, Y) = H_v^{15}(X, Y)$$

Final RGB Channel Evaluation

$$\text{Color}(x, y) = (F(H_0(X, Y)), F(H_1(X, Y)), F(H_2(X, Y)))$$

where $F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$ performs exact differentiable color clamping to $[0, 255]$.